

MEHEDI HASAN

Senior Full-Stack Engineer | AI Chatbot & Cloud Integration

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PROFESSIONAL SUMMARY

Full-stack Software Engineer specializing in enterprise-grade web applications, microservices, AI chatbot platforms, and cloud-integrated systems. Strong experience in .NET Core, React, Angular, Next.js, FastAPI, PostgreSQL, SQL Server, MongoDB, and AWS services. Skilled in building secure, scalable, multi-tenant applications with JWT/SSO authentication, role-based access control, streaming AI chat, knowledge retrieval, and automation workflows. Experienced in delivering business-critical systems for smart infrastructure, security management, ERP, eCommerce, and Town Council operations.

TECHNICAL SKILLS

Backend	C#, .NET Core, ASP.NET MVC, FastAPI, Python, Microservices, REST APIs, Entity Framework, Dapper
Frontend	React, Angular, Next.js 15, TypeScript, JavaScript, Tailwind CSS, jQuery, Bootstrap, HTML5, CSS3
Cloud & AI	Amazon Bedrock, Strands Agents, Amazon Lex, Knowledge Base, AWS Lambda, API Gateway, ECS/Fargate, CDK, CloudWatch
Database & Cache	SQL Server, PostgreSQL, MongoDB, DocumentDB, Redis, Valkey
Security	JWT, SSO, OAuth 2.0, OIDC, SAML, OpenIddict, RBAC, audit logging
Tools & DevOps	Git, Docker, PowerShell deployment scripts, OpenSSL, MQTTX, EMQX, MCP Servers, Visual Studio
Reporting & UI	RDLC, Crystal Reports, responsive UI, admin dashboards, embeddable iframe chatbot

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Xtroit

July 2025 - Present

- Built and maintained AI SMART Assistance, a secure multi-tenant AI chatbot platform for Town Council operations using Next.js, FastAPI, Amazon Bedrock, and Strands Agents.
- Implemented streaming AI chat responses, session management, chat history restoration, file upload, and tool execution progress tracking.
- Integrated MCP-based tools for knowledge base retrieval, database querying, case retrieval, PDF extraction, noticeboard access, and building maintenance workflows.
- Developed tenant-aware runtime configuration for tools, models, security settings, business rules, and user access control.
- Implemented JWT/SSO authentication, RBAC authorization, audit logging, and secure tenant/user context propagation.
- Worked on AWS deployment architecture using ECS/Fargate, Lambda, API Gateway, CDK, CloudWatch, and PowerShell automation.

Senior Software Engineer - Surbana Jurong Bangladesh Ltd.

May 2023 - June 2025

- Developed cloud-integrated smart infrastructure solutions using .NET Core, React, PostgreSQL, Docker, OpenIddict, MQTTX, EMQX, and microservice-based architecture.
- Contributed to the HDB Smart Lighting Platform for Singapore's Housing & Development Board, supporting smart lighting management, motion traffic statistics, and operational insights.
- Designed and reviewed backend services, APIs, and frontend modules to improve system maintainability and code quality.
- Worked on secure authentication, IoT data communication, and scalable service integration for Town Council and smart infrastructure use cases.

Senior Software Engineer - Masco Group

November 2016 - May 2023

- Led development of ERP, eCommerce, admin panel, and chat application features using .NET Core, ASP.NET MVC, Angular, React, PostgreSQL, and SQL Server.
- Developed garments ERP modules including merchandising, supply chain, knitting, reporting, and business workflow management.
- Built and maintained Masco Bazar, an online platform for daily commodity sales with admin management and customer-facing features.
- Created reports using RDLC and Crystal Reports and optimized database queries for business analytics and operational reporting.

Software Engineer - Advanced Software and IT Service

October 2015 - October 2016

- Developed CRM and ERP modules using ASP.NET MVC, C#, JavaScript, jQuery, and SQL Server.
- Worked on DealerCRM, supporting vehicle inventory management and customer lead tracking for car dealerships and finance providers.

KEY PROJECTS

AI SMART Assistance

- Developed a secure Town Council AI chatbot platform using Next.js 15, FastAPI, Amazon Bedrock, Strands Agents, MongoDB/DocumentDB, and AWS services. Implemented streaming AI chat, tenant-aware tool/model configuration, JWT/SSO authentication, MCP tool integrations, admin dashboards, chat history, file upload, and cloud deployment automation.

HDB Smart Lighting Platform

- Developed and maintained a centralized smart lighting management platform for HDB and Town Councils to manage vendor-based smart lighting systems, motion traffic statistics, heat map metrics, and maintenance planning support.
- Tech stack: React, .NET Core 6, Microservices, OpenIddict, PostgreSQL, MQTTX, EMQX, OpenSSL, Docker

KONDOLINK

- Worked on a web-based security management platform for building operations, supporting clients including SMS Security Canada and MAGNUM Security USA.
- Tech stack: .NET Core, Angular, SQL Server

ERP & eCommerce Systems

- Developed ERP, eCommerce, admin panel, reporting, and chat application features for Masco Group and Masco Bazar.
- Tech stack: .NET Core, ASP.NET MVC, Angular, React, PostgreSQL, SQL Server, RDLC, Crystal Reports

EDUCATION

M.Sc. in Computer Science - Jahangirnagar University, 2017 | CGPA: 3.35/4.00

B.Sc. in Computer Science and Software Engineering (CSSE) - American International University-Bangladesh (AIUB), 2015 | CGPA: 3.53/4.00